60-second research explanation

133 words | Approximate delivery: 130 words/minute. Rehearse and adjust pauses.

An X-ray and a patient's pain score can tell different stories. We asked whether that mismatch helps identify knee pain predictions that will be less reliable four years later. We analyzed 8,232 knees from 4,146 Osteoarthritis Initiative participants, keeping each person's knees together during validation. The unadjusted analysis linked greater mismatch with larger prediction errors. But an important follow-up changed the interpretation: after adjusting for baseline pain, mismatch no longer had an independent association with error. Baseline symptom burden was the stronger explanation. We also tested whether an uncertainty model could defer difficult cases. Setting aside the most uncertain ten percent lowered error among retained cases by about eleven percent, but coverage fell and deferral varied by subgroup. This is internally validated research, not a clinical tool. The next step is external validation.

Prepared from the supplied manuscript and frozen aggregate outputs. No completed presentation or award is claimed.

Three-minute research explanation

395 words | Approximate delivery: 140-150 words/minute. Rehearse and adjust pauses.

Imagine two knees with similar X-rays but very different levels of pain. That disconnect is common in osteoarthritis. Our question was whether this structure-pain mismatch identifies cases whose future pain will be harder for a model to predict.

We conducted a secondary analysis of the Osteoarthritis Initiative. The baseline cohort contained 8,920 knees. Our primary analysis used the 8,232 knees from 4,146 participants with observed pain scores at 48 months. The outcome was WOMAC pain, measured on a zero-to-twenty scale.

We defined discordance as the absolute difference between reported baseline pain and expected pain for the same X-ray severity grade. Crucially, those expectations were estimated inside training folds. We used elastic-net regression with nested five-fold cross-validation grouped by participant. Both knees from one person stayed together, and preprocessing and tuning were learned without accessing the outer test outcomes.

The model's held-out mean absolute error was 1.787 points. In the frozen unadjusted primary analysis, each additional point of discordance was associated with about 0.301 points of additional absolute prediction error.

But that is not the whole finding. Discordance was strongly correlated with baseline pain. In a post-freeze sensitivity analysis, adjusting for baseline pain reduced the discordance coefficient to minus 0.011, with a confidence interval crossing zero. So we cannot claim that discordance independently explained prediction failure. Baseline symptom burden was the stronger candidate explanation in these data. That qualification is central to interpreting the project.

We also studied selective prediction: could a model recognize cases where its predictions were likely to be less reliable? A separate uncertainty model learned from inner out-of-fold errors and scored untouched outer-test cases. Deferring the most uncertain ten percent retained 7,409 knees and lowered their mean absolute error to 1.582, about an eleven percent relative reduction. That benefit comes with reduced coverage. Deferral rates varied by subgroup, and because the uncertainty model included discordance, its performance is not independent confirmation of a discordance effect.

This study used one cohort and internal validation. It does not establish causation, clinical benefit, or transportability to new populations. The baseline-pain adjustment was post hoc, and the public interactive explorer uses clearly labeled fictional examples rather than a patient-facing prediction model.

The broader lesson is that evaluating a model means asking not only how accurate it is, but also what explains its errors and which cases it leaves out. External validation is the next scientific step.